

UNSUPERVISED IDENTIFICATION OF EVENTS IN CHICAGO AIR-QUALITY DATA

Feature Importance and K-Means Clustering of Fine Particulates, Gases, and Weather

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MOTIVATIONS

- Air quality data typically consists of particulate matter ($PM_{2.5}$, $PM_{1.0}$, and $PM_{10.0}$) and other gases (CO , NO , NO_2 , and O_3). Monitoring PM and gas percentages near the surface helps determine the safety and quality of the air for those looking to be situated outside
- Summer 2023 saw multiple Canadian wildfire plumes impacting Northeastern Illinois. These plumes carried some of the worst air quality seen in the region to date
- Therefore, classifying the periods based on the air quality measurements is necessary



Per-Cluster Variable Treemaps

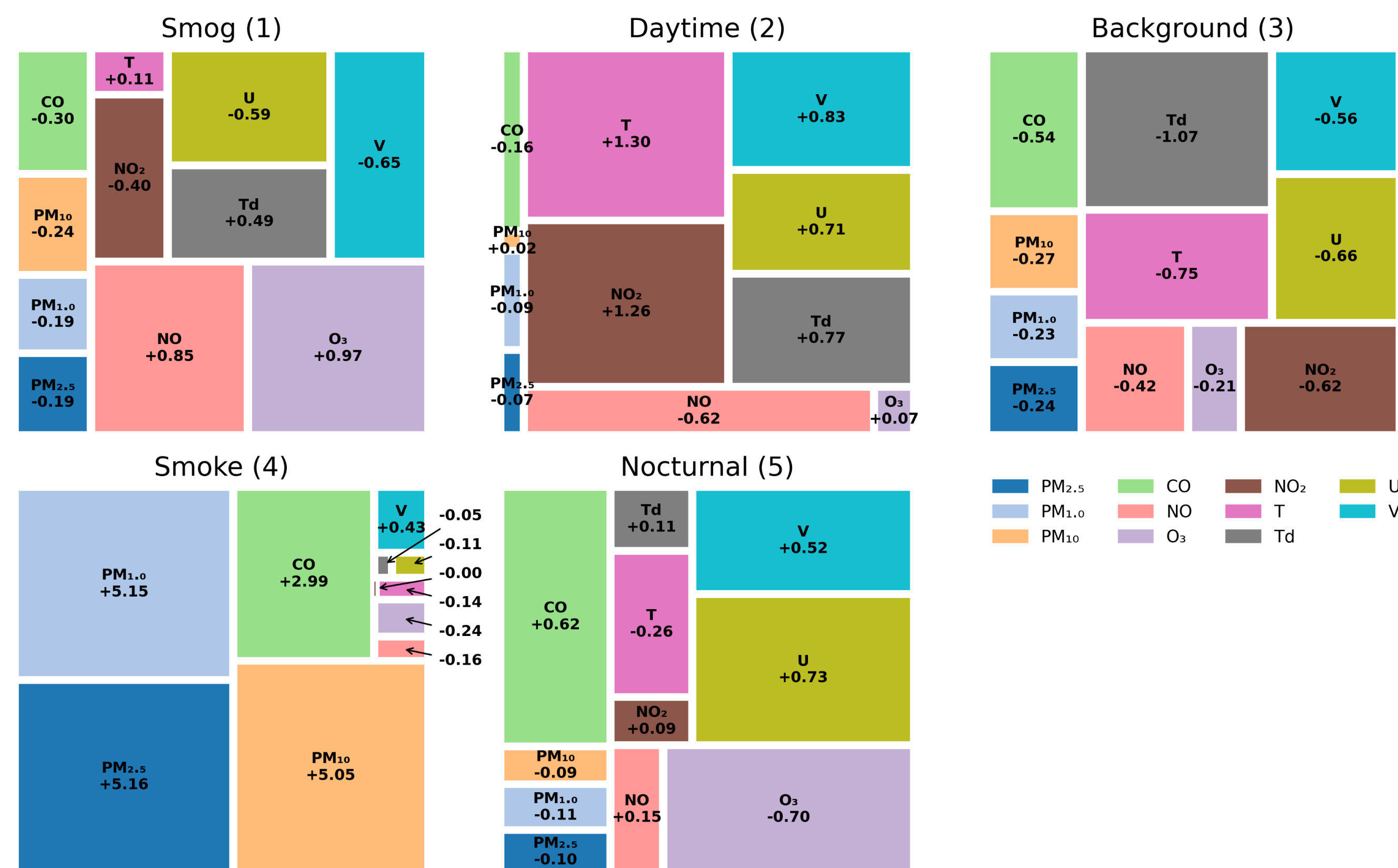


Figure 1: Variable treemaps for each of the five clusters, where the tile area reflects the standardized deviation of each pollutant or meteorological variable from its overall mean. The negative values are below the standard deviation and positive values are above the standard deviation

Boxplots of Each Variable by Cluster

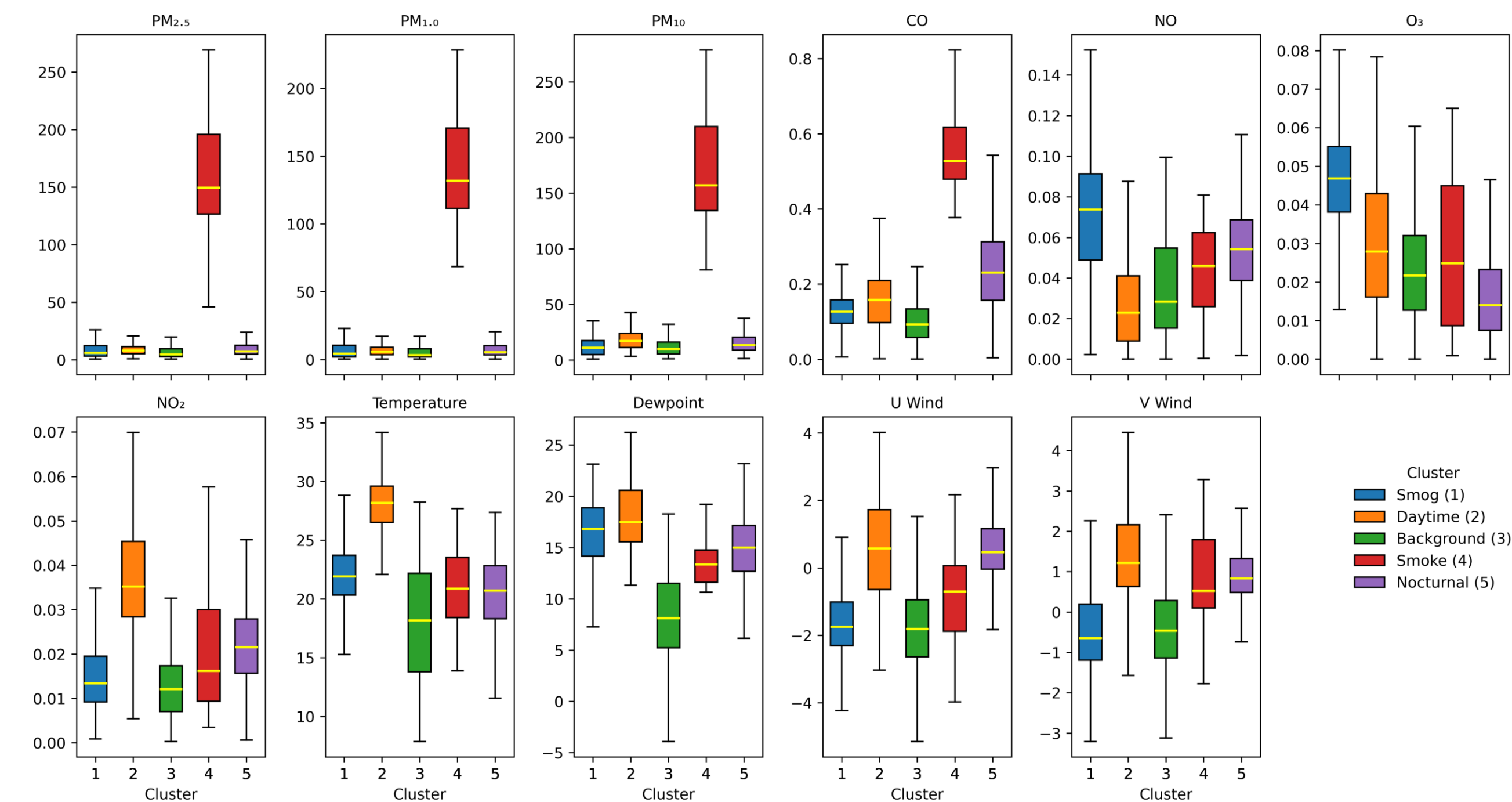


Figure 3: Boxplots of air quality and meteorological variables by cluster, illustrating distributions in particulate matter, gases, temperature, dewpoint, and wind components. Each cluster's unique signature – ranging from intense smoke and smog to nocturnal and background conditions – is clearly differentiated by the position and spread of the boxes

OVERVIEW

- Applied K-means clustering to 10-minute rooftop measurements ($PM_{1.0}$, $PM_{2.5}$, PM_{10} , CO , NO , NO_2 , O_3 , temperature, dewpoint, U/V wind) at NEIU from May through August 2023
- Identified five regimes: Smog, Daytime, Background (Stable low-pollutant conditions), Smoke (Wildfire influence with moderate to high PM/CO), and Nocturnal
- Standardized treemaps, diurnal frequency plots, and time-series highlighting smoke intervals were used to separate local emissions, regional background, and meteorological effects

METHODS

- Air quality and weather sensor data from the NEIU CROCUS site was used from the Summer of 2023
- The study used the Vaisala AQT (air quality) and WXT (weather) sensors to measure particulate matter and gaseous pollutants, plus meteorological parameters: temperature, dew point, humidity, rain rate, wind speed, and direction
- Preprocessing was done to mask invalid data, the data was then averaged to 30 minutes, and then standardize the data through zero mean and standard deviation
- K-means clustering – which is an unsupervised algorithm that splits the data into k groups, assigning each point to whichever cluster centroid (mean) it's closest to – and then choosing the number of clusters (k) based on the elbow plot and by eye
- Various plots were created to determine cluster quality and to determine features within clusters: histograms, boxplots, pairplots, and treemaps of cluster centroids (Figure 1)

Overview of Cluster Statistics

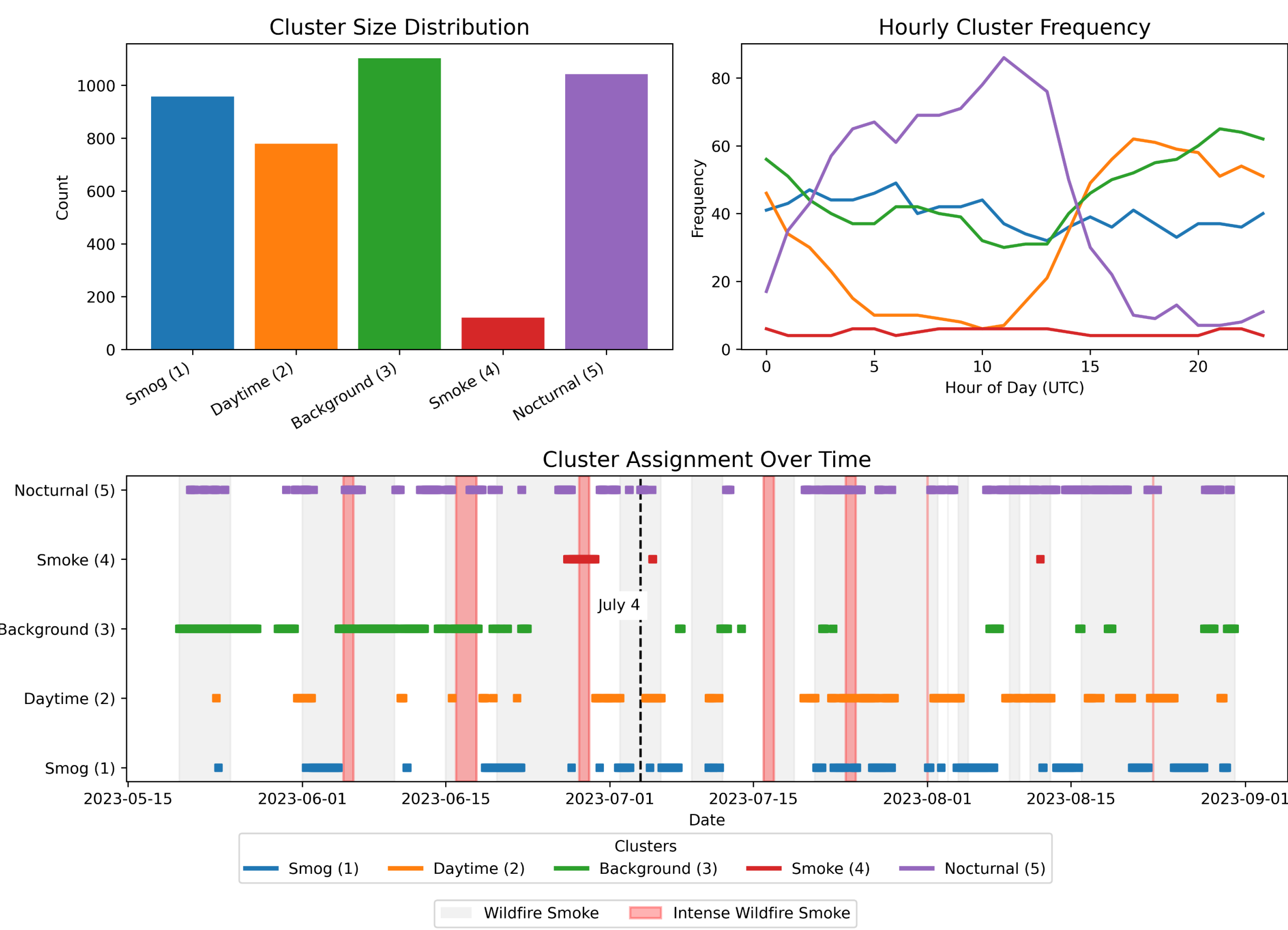


Figure 2: This composite view summarizes the five clusters: the bar chart (top left) shows each cluster's relative frequency; the line plot (top right) displays their average hourly frequency (UTC), highlighting smog, daytime, smoke, background, and nighttime signatures; and the timeline (bottom) displays cluster assignments over May 20, 2023 through August 30, 2023, with light gray shading marking wildfire smoke periods, red bands indicating intense smoke events, and a dashed line on July 4 for fireworks

RESULTS

- Patterns in pollutant levels (Figure 2) streamlined cluster identification by highlighting distinct temporal and other signatures
- Cluster 4 (Smoke) was characterized by high PM and CO levels
- The frequency time plot assists in determining the diurnal aspect of clusters
- Per-Cluster variable treemaps assisted in determining the ambiguous clusters (Figure 3)
- Smog (Cluster 1) was determined from the high NO and O_3 compared to the other variables
- Background (Cluster 3) showed uniformly low values across all variables

CONCLUSIONS

- The clustering separates the sustained multi-day haze of wildfire smoke – characterized by persistently high PM and CO – from the short, sharp spike in particulates around July 4th due to fireworks
- Daily cycles reveal midday peaks in O_3 (Ozone) under strong sunlight alongside pronounced morning/evening NO/CO surges from traffic, all riding on a steady urban “background” baseline
- Open-source dataset of air quality in Chicago for May through August 2023

NEXT STEPS

- Determine if the Planetary Boundary Layer (PBL) height and other parameters has any effect on air quality and surface conditions
- Compare air quality data with meteorological conditions to determine potential synoptic scale drivers for air quality

REFERENCES

O'Brien J ; Anderson G ; Muradyan P ; Tuftedal M ; Collis S ; Jackson R ; Grover M ; Dematties D ; Park S ; Sankaran R ; Beckman P ; Ferrier N ; Kim Y ; Shahkarami S (2024): CROCUS Air Quality Data at Northeastern Illinois University Rooftop. Community Research on Climate and Urban Science Urban Integrated Field Laboratory (CROCUS UIFL), ESS-DIVE repository. Dataset. doi:10.15485/2335799 accessed via <https://data.ess-dive.lbl.gov/datasets/doi:10.15485/2335799> on 2025-07-17

O'Brien J ; Anderson G ; Tuftedal M ; Grover M ; Muradyan P ; Collis S ; Park S ; Dematties D ; Sankaran R ; Ferrier N ; Shahkarami S ; Kim Y ; Keeler E (2024): CROCUS Weather Data at Northeastern Illinois University Rooftop. Community Research on Climate and Urban Science Urban Integrated Field Laboratory (CROCUS UIFL), ESS-DIVE repository. Dataset. doi:10.15485/2338027 accessed via <https://data.ess-dive.lbl.gov/datasets/doi:10.15485/2338027> on 2025-07-17